

LDV_CNG

Compressed Natural Gas Light Duty Vehicle



CNG Car

Inputs / Outputs

dir	comm	unit	group
Input	CNG	PJ	i
Output	PLDVHWY	MPKm	o
Output	PLDVCTY	MPKm	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
2	PLDVHWY	707.214	2.03	0.4
3	PLDVCTY	522.466	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
stock	849.80 1000 Vehicles
cap unit	1000 Vehicles
act unit	million km, city
cost unit	USD/(Vh*km)

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	CNG	3.95	2.55	1.41	59.55
City	CNG	5.35	1.88	1.91	44.00

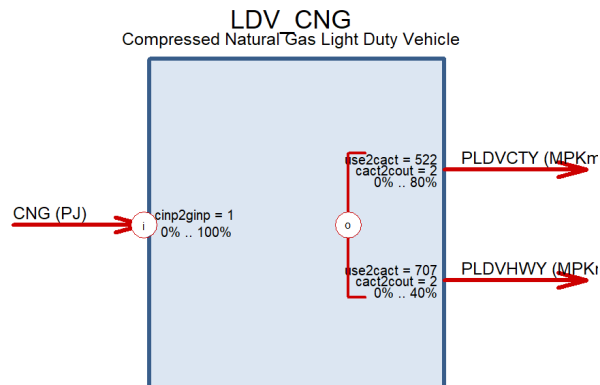
Vehicle Parameters

Fuels	CNG	Primary fuel	CNG
Efficiency (highway)	1.414 MJ/km	Efficiency (city)	1.914 MJ/km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Vehicle cost	7,677 USD	Annual fixed cost	25 USD/yr
Lifetime	16 yr		

Capacity bounds

region	year	stock	cap.lo	cap.up	cap.fx	ncap.lo	ncap.up	ncap.fx	ret.lo	ret.up	ret.fx
IND	2024	849.8	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2025	796.7	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2030	531.1	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2035	265.6	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2040	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2045	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2050	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA

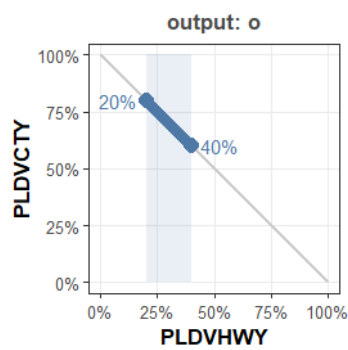
Schematic



Costs [USD/(Vh*km)]

Fixed O&M	0.02
Investment cost	7.68

Share Mix Constraints



Levelized Cost 0.13 [USD/(Vh*km)], 0.06–0.07 [USD/(Passenger*km)]

